

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety, Tolerability and Pharmacokinetics of AZD5148 in Healthy Adults **Short Title/ Acronym:** AZD5148 Phase I Healthy Adults Study
Protocol Number: D8820C00002 **NCT Number:** NCT06469151 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD5148 (anti-TcdB mAb) **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To evaluate the safety and tolerability of a single dose of AZD5148 administered via intravenous (IV) bolus or intramuscular (IM) injection.
Secondary Objectives: To evaluate the pharmacokinetics (PK) of a single dose of AZD5148.

2.2 Background & Rationale
To determine the safety, tolerability, and PK profile of AZD5148 in healthy adults. AZD5148 is a monoclonal antibody targeting *Clostridioides difficile* toxin B (TcdB) in development for the prevention or reduction of symptomatic *C. difficile* infection (CDI) and its recurrence. This first-in-human study provides foundational safety and dosing data before advancing the therapy to patient populations.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional **Control Type:** Placebo-controlled
Blinding: Double-blind **Randomization:** Randomized (dose-escalation cohorts, with a 9:1 ratio for main groups after Sentinel dosing)

3.2 Planned Enrollment & Duration
Target \$N\$: 84 participants **Number of Sites:** Four clinical units
Estimated Study Start: 24-Jun-2024 **Estimated Primary Completion:** 22-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria
1. Healthy participants, aged 18 to 65 years, with a Body Mass Index (BMI) ≥ 18.0 to ≤ 32.0 kg/m² and body weight $\geq$

45\$ kg to \$\\le 110\$ kg. 2. Suitable veins for cannulation or repeated venipuncture at the time of consent. 3. Women of childbearing potential must have negative pregnancy tests, must not be lactating, and if heterosexually active, must agree to use an approved method of highly effective contraception from 3 months prior to administration until 360 days after the dose.

4.2 Exclusion Criteria

1. History of any clinically important disease, disorder, medical/surgical procedure, or trauma within 4 weeks of the first administration of the study drug. 2. History of malignancy other than treated non-melanoma skin cancers or locally treated cervical cancer in previous 5 years. 3. Any medical history of symptomatic CDI within the prior 2 years. 4. Previous receipt of a mAb within 6 months, or 5 antibody half-lives (whichever is longer), prior to the start of the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: Single, discrete dose of AZD5148 or matching placebo (dose escalation across cohorts: e.g., dose 1 i.m., dose 2 i.m., dose 2 i.v.). **Route of Administration:** Intramuscular (IM) injection into the lateral thigh muscle or Intravenous (IV) bolus. **Duration of Treatment:** Single administration followed by a 12-month follow-up period.

5.2 Concomitant & Prohibited Therapy

Permitted: Standard medications permitted for healthy volunteers that do not interfere with PK/safety evaluation, at the investigator's discretion.

Prohibited: Receipt of immunoglobulin or blood products within 6 months prior to screening or expected to be received during the study; systemic glucocorticoid therapy exceeding 2 weeks of prednisone or equivalent within 6 months prior to screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to Day -1	Informed Consent, Medical History, Viral Serology (HBsAg, HCV, HIV), Pregnancy Tests, Baseline Labs
Baseline / Dosing	Day -1 to Day 2	Clinical Unit residency, Randomization, Single Dose Administration (Sentinel monitoring of 24h before full cohort), Safety PK sampling
Interim	Weeks 1 to 51	Outpatient follow-up visits for Safety and PK evaluation
End of Study	Day 361 $\pm$ 14	Final Follow-up Visit: Final Vital Signs, Exit Interview, Safety & PK Check

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint

Definition: Safety and tolerability evaluated by the incidence of adverse events (AEs), serious adverse events (SAEs), and vital sign/laboratory abnormalities. **Measurement Method:** Clinical observation, physical examinations, and central/local laboratory analysis.

7.2 Secondary & Exploratory Endpoints

1. Pharmacokinetic (PK) profile of a single dose of AZD5148 (including C_{max} and AUC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection during residency at the Clinical Unit (Day -1 to Day 2) and throughout the 12-month outpatient follow-up period. **Serious Adverse Events (SAEs):** Reported to the sponsor typically within 24 hours of site awareness.

Data Monitoring Committee (DMC): Internal safety monitoring utilized to assess Sentinel Group safety data (24-hour observation) before advancing to the remaining participants in each cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all randomized subjects receiving at least one dose of the investigational product).

Sample Size Justification: $N = 84$ healthy volunteers is considered adequate to determine the initial safety, tolerability, and PK parameter characteristics for the ascending single doses in a Phase I First-In-Human setting.

Handling of Missing Data: Standard approaches for Phase I trials (e.g., participants with missing critical PK data are typically excluded from the specific PK analysis set).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical monitoring visits to verify source data (SDV), protocol adherence, and investigational product accountability.

Audit Trail: Complete eCRF locking processes, verification of clinical unit standard operating procedures, and automated data clarification forms.

11. Study Identifiers & Governance

Primary ID: D8820C00002 **Registry Number:** NCT06469151 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study to Evaluate the Safety, Tolerability and Pharmacokinetics of AZD5148 in Healthy Adults **Official Regulatory Title:** A Phase I Double-Blind, Placebo-Controlled Study to Evaluate the Safety, Tolerability and Pharmacokinetics of AZD5148 in Healthy Adults

12. Clinical Framework (AZD5148 / CDI Specific)

Primary Condition: Healthy Volunteers (For *Clostridioides difficile* infection development program) **Diagnosis Threshold:** N/A (Healthy adults) **Medical Methodology:** Anti-TcdB monoclonal antibody administration **Optimization Target:** Characterization of single-dose safety, tolerability, and PK

13. Study Procedures & Methodology

Management Pathway: Phase I single ascending dose (SAD) escalation **Education Protocol (HCP):** Site training on protocol criteria, IM vs. IV administration techniques, and intensive PK sampling guidelines.

Education Protocol (Patient): Informed consent detailing prolonged highly-effective contraception requirements (360 days post-dose), study residency restrictions, and AE reporting.

Quality Audit Mechanism: Site monitoring and internal safety review during Sentinel dosing stages.

14. Observation & Follow-Up Schedule

Total Duration: Up to 56 weeks per participant (including 28-day screening and ~52-week follow up) **Onsite Visit Cadence:** Resident at the Clinical Unit from Day -1 until all assessments are completed on Day 2 **Remote/Outpatient Monitoring Frequency:**

Periodic ambulatory follow-up over 12 months

Checklist Review Interval: Final evaluation completed on Day 361
\$\\pm\$ 14 post-administration

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				